THE COTANGENT COMPLEX OF A MAP OF PERFECT RINGS IN CHARACTERISTIC P VANISHES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement is two sentences; the proof is a two-line argument combining an isomorphism with a null homotopy.

Editorial note. The single reference of the counted spans is recorded as a non-load-bearing navigation pointer; the argument that consumes it stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- The Frobenius-induced endomorphism is null homotopic in characteristic p. Lemma lemma-frobenius-acts-as-zero (2017-2023, produced as L44163), used at proof lines 2043-2045. Shows the second of the two properties of the endomorphism.
- Perfection of a ring makes Frobenius an isomorphism. used at proof lines 2043-2044. Makes the induced endomorphism an isomorphism.
- The functoriality of the cotangent complex. Section section-functoriality, used at proof lines 2043-2045. Supplies the induced endomorphism whose two properties are compared.

Source reference. Original lemma 0G5Z: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20179–20179. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let p be a prime number. Let $A \to B$ be a ring homomorphism and assume that $p = 0$ in A. If A and B are perfect, then $L_{B/A}$ is zero in $D(B)$.*

Proof. The map $(F_A, F_B) : (A \to B) \to (A \to B)$ is an isomorphism hence induces an isomorphism on $L_{B/A}$ and on the other hand induces zero on $L_{B/A}$ by Lemma 0G5Z. □

Date: stacks-project:cotangent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.